Setting the LS machine's patch size on the RHM

The length scale is the whole story -- and it must be chosen per site, not per step

The criterion "largest window matching any training patch" beats the best FIXED window in 12/12 cells, median 12x.

Every floor on top of it loses: the entropy floor h_min in 12/12 cells, and its count rival n_min at both P rungs. Rejection only shortens.

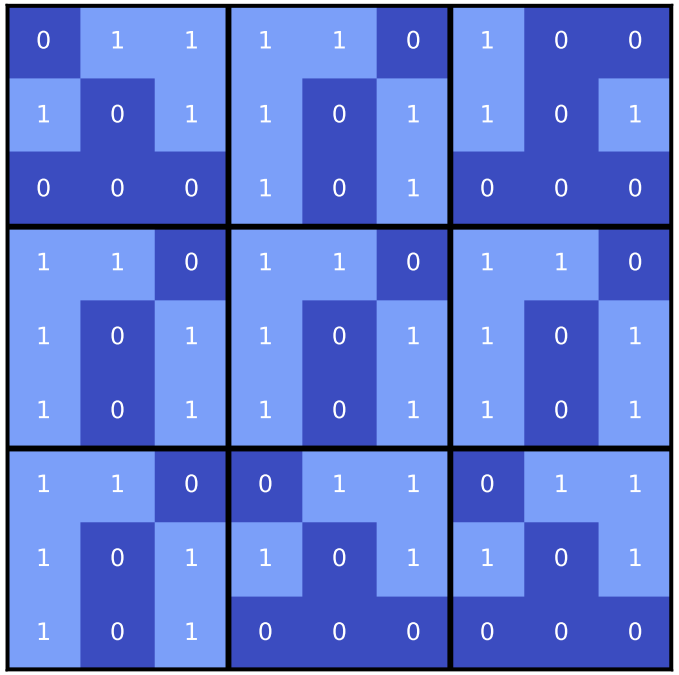
Replacing the per-site choice by its own induced k(t) schedule, at the SAME mean k, keeps only 15% of the success (9/9 cells).

Position explains none of the choice ($R^2 < 0.001$); local reveal count explains at most 0.47. The rest is content, and that content is grammar.

The block's own ambiguity adds +0.107 R^2, and its consistency with the OTHER blocks through the root -- only a straddling window sees it -- +0.196.

Tokenization: what a "sample" is, and what a window sees

one sample: 9x9 grid of tokens in {0..1}
black lines bound the 3x3 level-1 blocks



Tokens. Every site holds one of v = 2 symbols. There is no intensity and no ordering: token 2 is not "between" 1 and 3. One grid is 9x9 = 81 categorical sites.

Grammar. A root symbol expands L = 2 times; each symbol is replaced by one of m synonymous 3x3 "words", drawn once and then fixed. A grid is grammatical iff some root derives it. |R_L| = v*m^(1+s^2), so m^10 at s=3.

Diffusion. Absorbing-state ("erasure"): each site is independently replaced by a MASK token with probability p(t). Unmasked sites are never corrupted, so exact Bayes denoising IS patch matching -- no score approximation enters, and any gap from optimal is the window choice, not a fitting error.

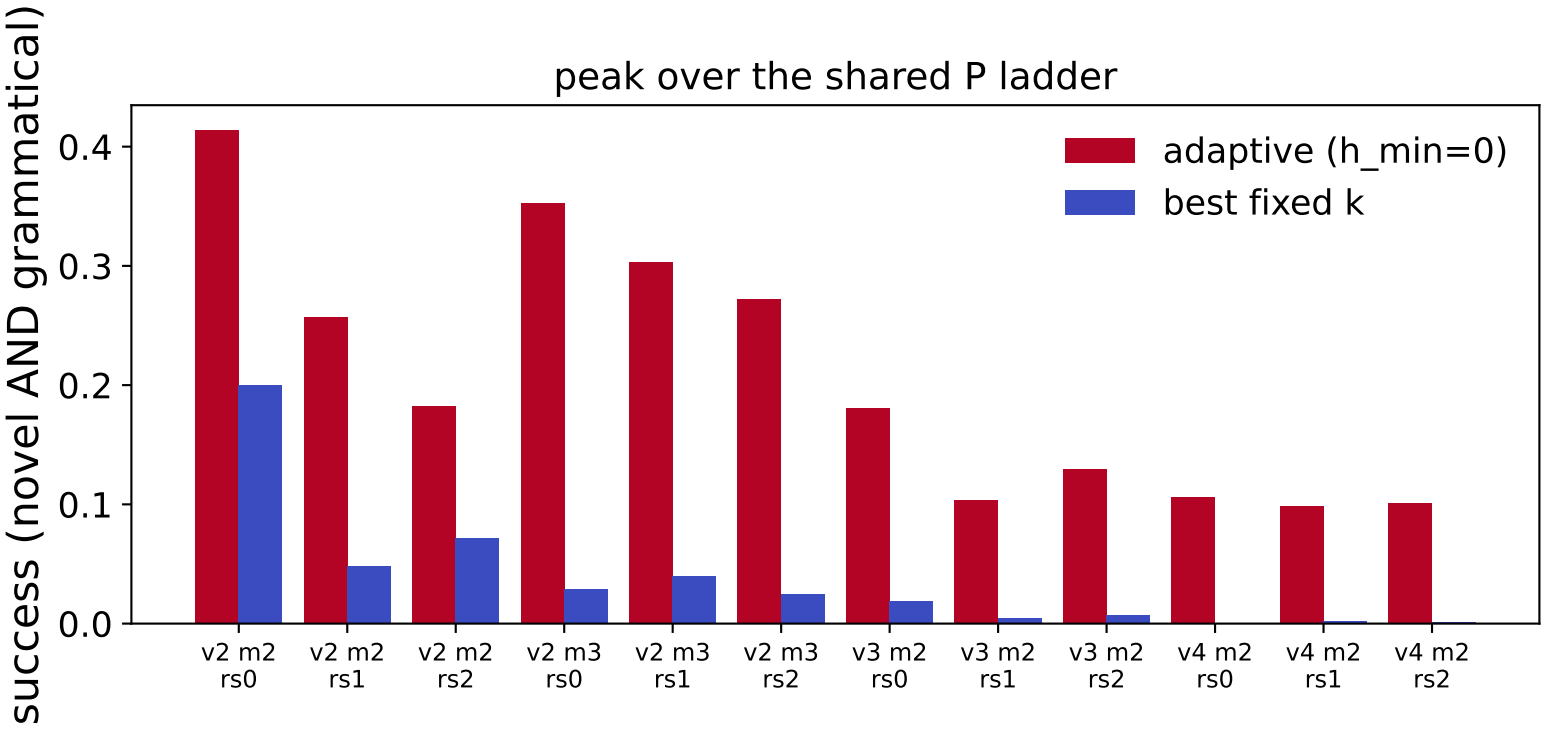
LS machine. The posterior at a masked site is the empirical distribution of the centre tokens of all training kxk patches agreeing with the observed sites in that window (Laplace alpha=1e-3). Sites are revealed one at a time.

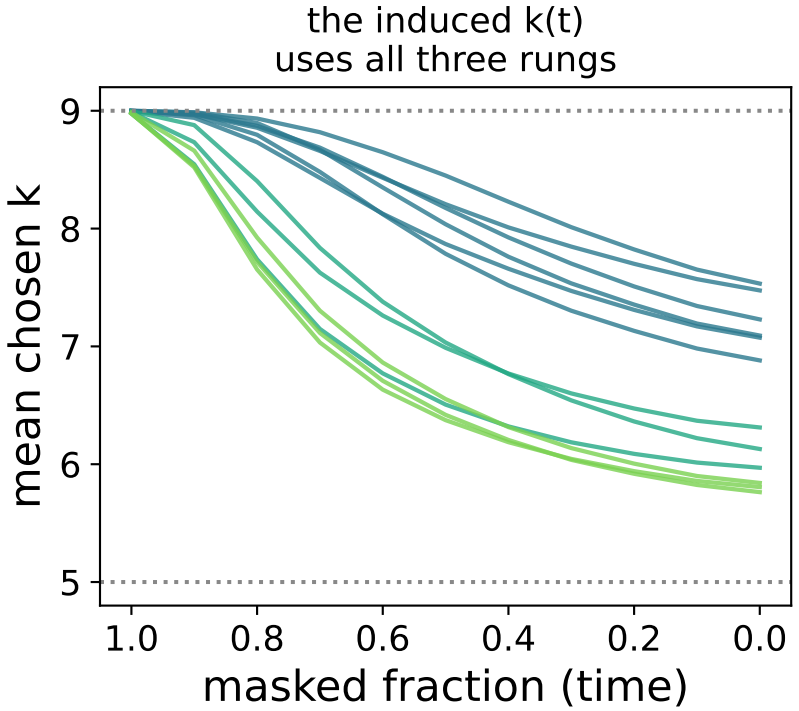
Why s=3. U1 = 2s^(L-1)-1 is the smallest window that can contain a whole top-level child block. At s=2, U1 = side-1, so the only odd rung in [U1, side] is U1 itself and there is no schedule to choose. At s=3, L=2 the board is 9x9 with three real rungs: 5, 7, 9.

Success (hshunt's criterion): a grid counts only if it is NOT in the training set AND parses completely.

success = grammatical AND novel

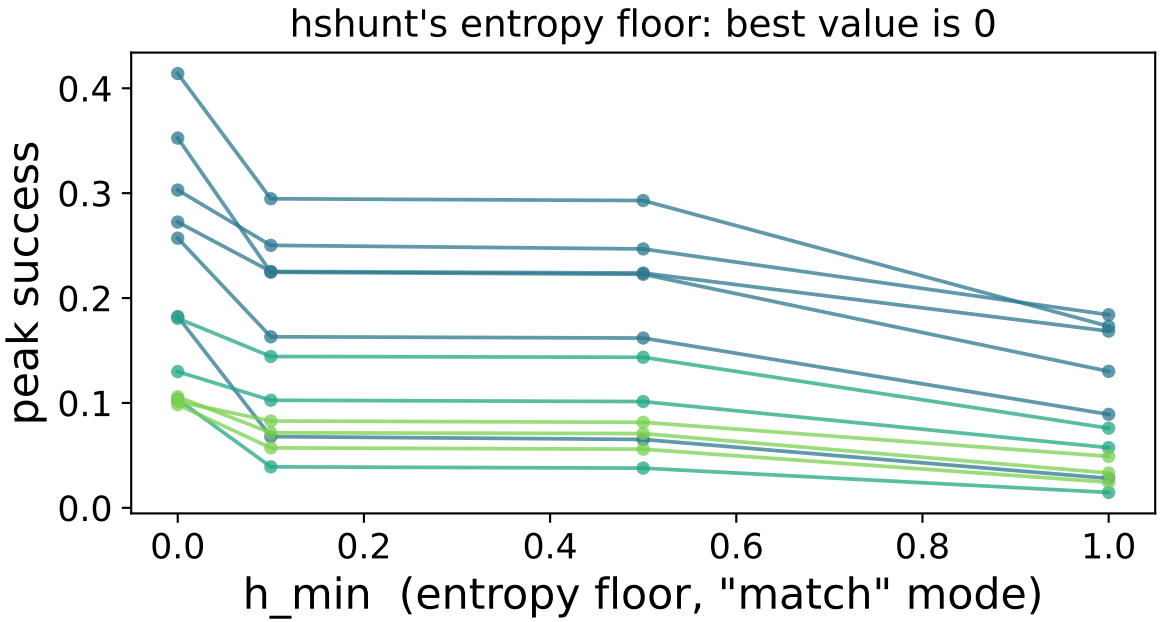
1. The window choice matters: adaptive beats every fixed k

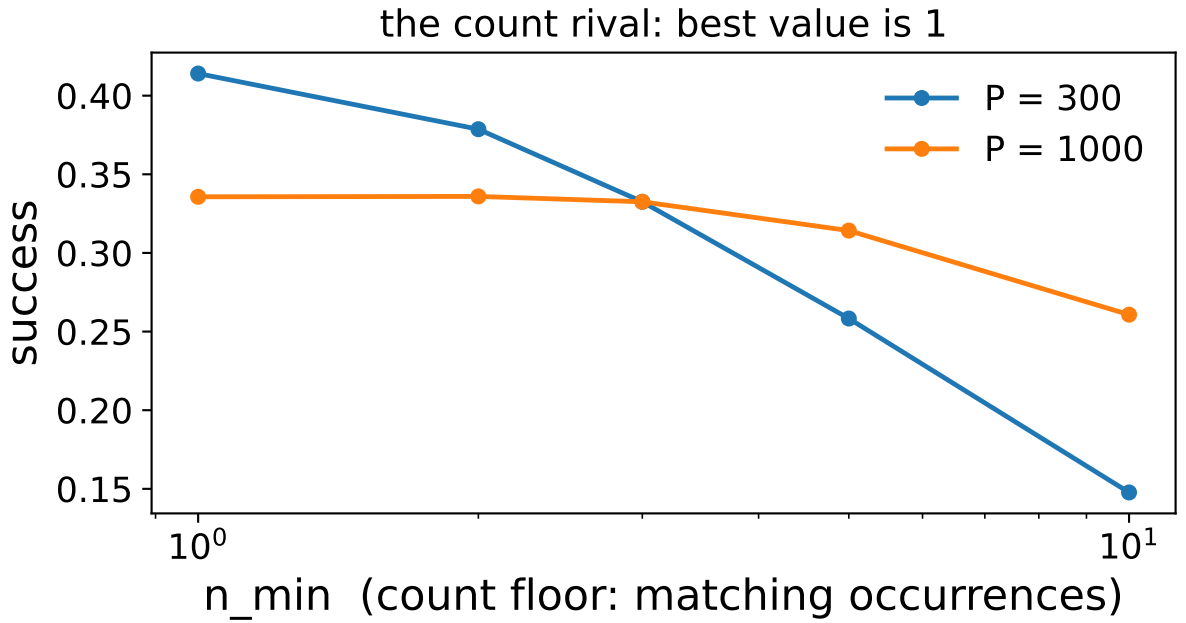





```
Adaptive beats the best fixed k in 12 of 12 cells; median ratio 12x.
The fixed optimum sits at U1 = 5 in only 2 of 12 -- it prefers k = 7, so the
machine wants a window LARGER than the smallest one that can hold a child block.
k(t) falls from 9 as the grid fills, and its span grows with the vocabulary v
(about 1.5 at v=2, 2.9 at v=3, 3.2 at v=4): a richer grammar needs to back off
further as the grid fills.
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2. Neither floor earns its keep -- the bare support test wins



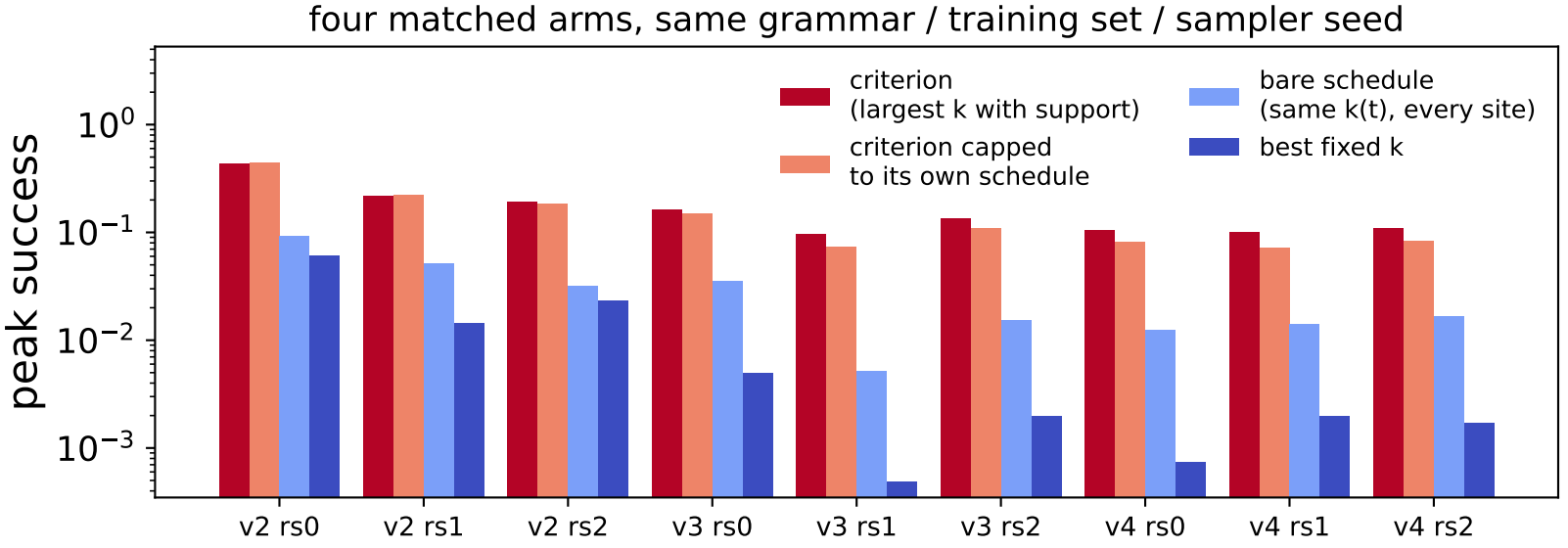


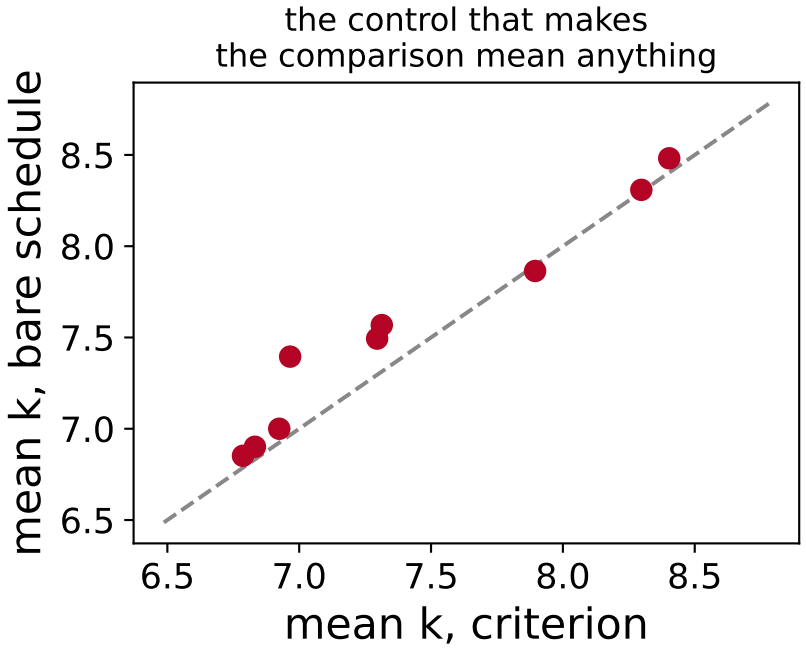

```
The machine accepts a window when  n_match > 0  AND  h >= h_min.
In "match" mode h = log(effective number of matching training patches), so
h_min is a support-DIVERSITY floor and h_min = 0 is bare support-EXISTENCE.

n_min is the genuine rival, not a reparametrisation: n_match counts matching
training OCCURRENCES, h_match measures their DIVERSITY. A window matched by
fifty copies of one patch has n_match = 50 but h_match = 0 -- the entropy floor
throws it away, a count floor keeps it. Both lose anyway.

One mechanism explains both: rejecting a window can only send the site to a
SHORTER one, and shorter is less grammatical. The floors buy novelty and sell
grammaticality at a bad rate (n_min 1 -> 10: novel +0.14, grammatical -0.40).
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3. Nor is it a schedule: the SAME k(t), applied per site, is worth 4.7x



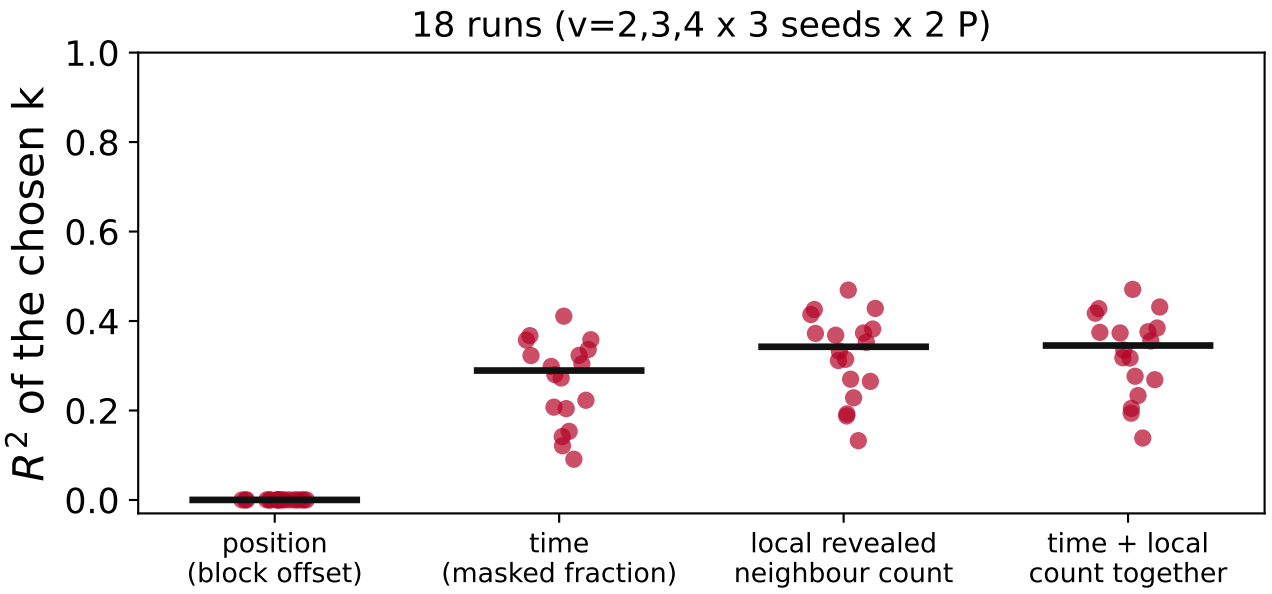


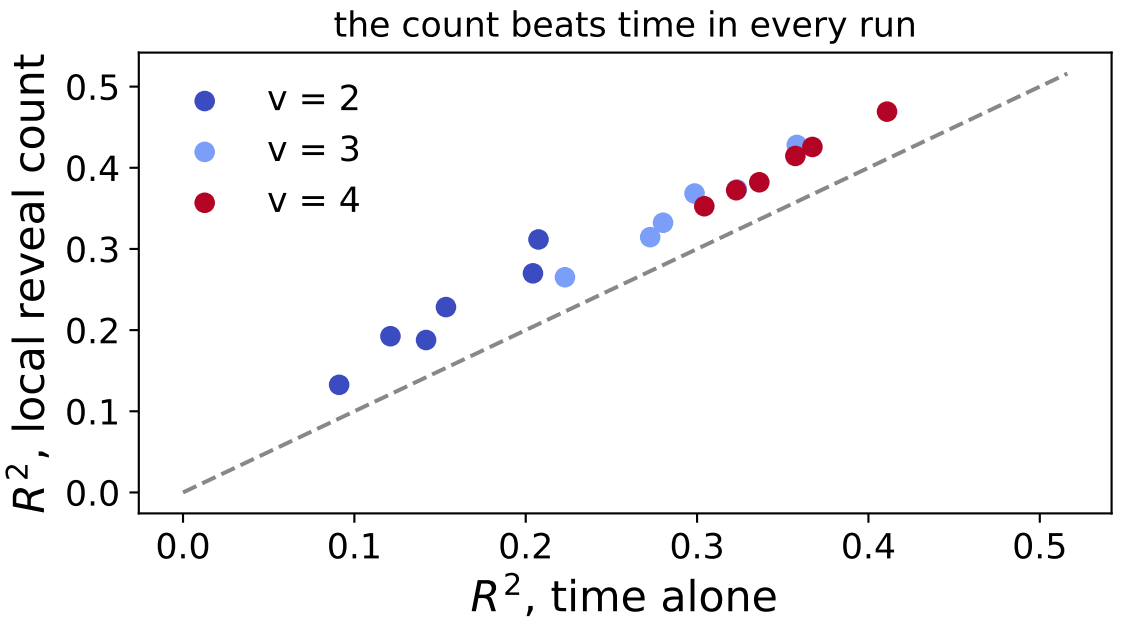
The control schedule is derived from the adaptive arm's OWN recorded trace -- round its mean k(t) to the available rungs -- so it cannot be rigged by picking a schedule that was always going to lose.

Adaptive beats the bare schedule in 9/9 cells and the best fixed k in 9/9. The schedule recovers a median 15% of adaptive success WHILE SPENDING THE SAME WINDOW BUDGET: median gap in mean k is 0.08 (right panel, on the diagonal).

So the win is not which window sizes get used on average. It is which SITE gets which size. Capping the criterion to its own schedule is free at v=2 but costs 8-30% at v=3,4, so the per-site freedom matters more as the grammar grows.

4. What the per-site choice actually depends on



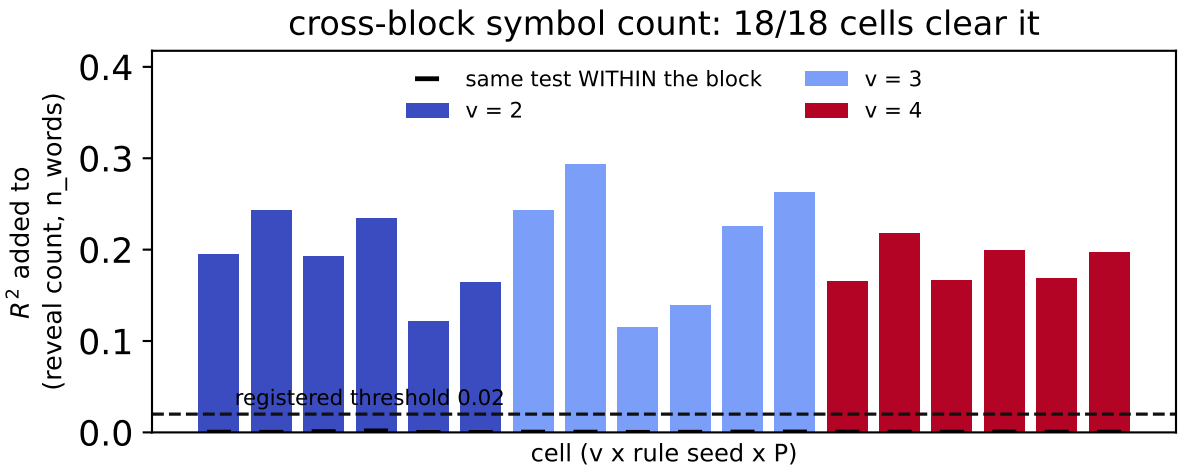


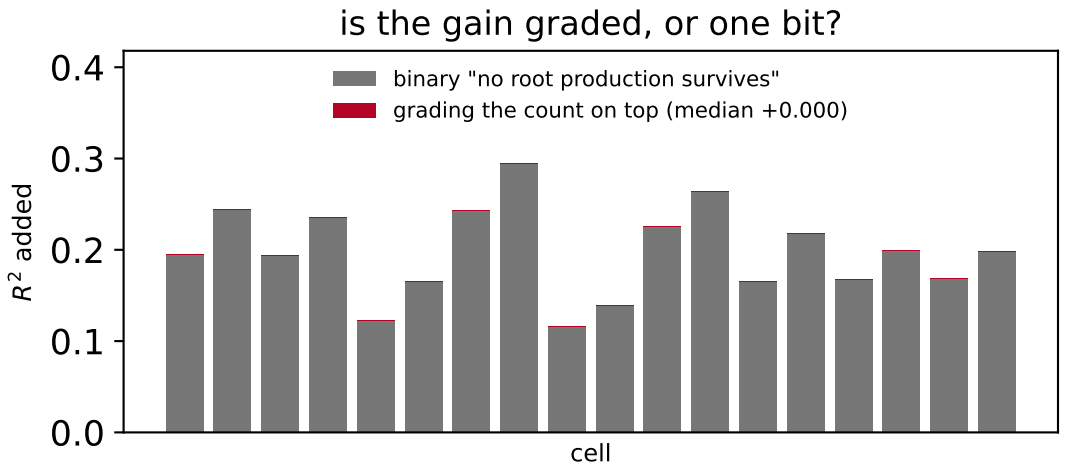
Position explains nothing. Regressing the chosen k on the site's offset within its level-1 block gives R^2 < 0.001 in all 18 runs, and adds nothing jointly with time. So no lookup table in (position, time) can replace the criterion -- and the window choice carries no signature of how it straddles the grammar blocks.

The local revealed-neighbour count (how many of the 9x9 around a site are fixed) beats time in 18/18 runs and SUBSUMES it: adding time on top gains only .005-.013. The k(t) schedule is therefore a consequence of reveal density, not a separate effect -- which is why page 3's bare schedule fails at matched mean k.

But it caps at 0.47. The remaining >= 53% turns on WHICH tokens were revealed, not how many. The criterion is irreducibly content-dependent.

5. The content residual is grammar consistency ACROSS blocks





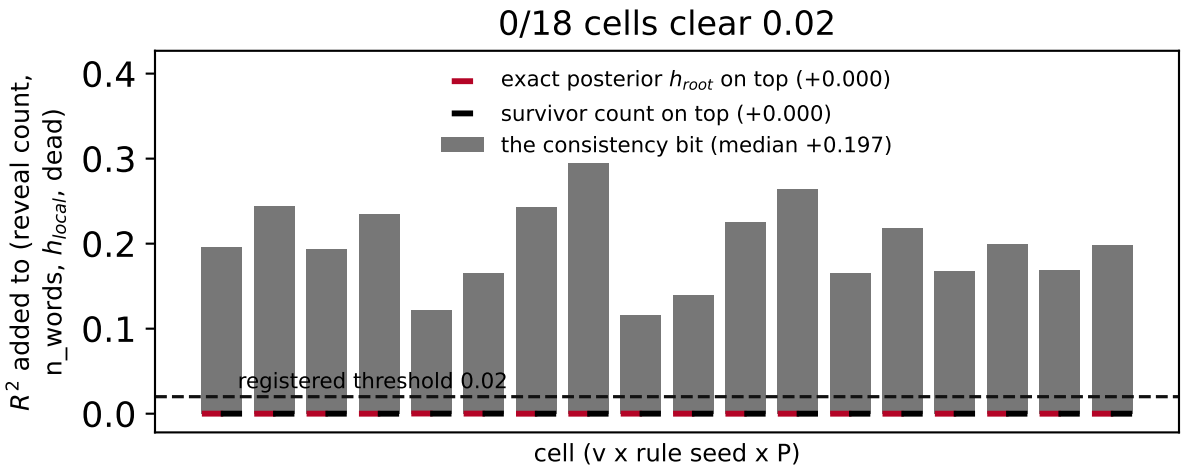
A site's own level-1 block cannot tell whether it is consistent with the OTHER blocks. Only a window straddling a boundary can. n_sym_root is that test made countable: how many symbols the block still admits AFTER every other block has constrained it through the root production.

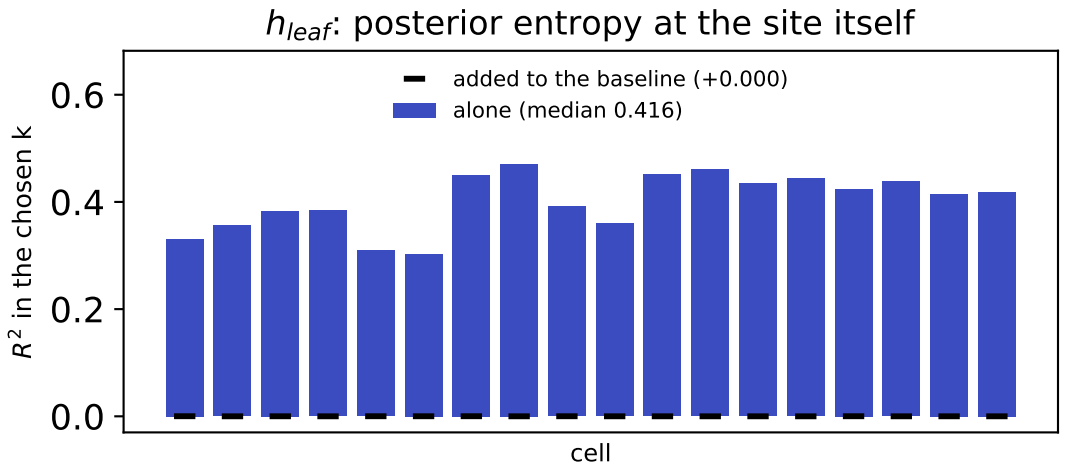
The substrate is there: the root tightens the block at 46-71% of observations, rising with v. And the criterion uses it. Added to (reveal count, n_words) it gains a median +0.196 R^2, in 18/18 cells, against a permutation null of +0.000 -- the count shuffled within each step, which measures how much any fourth column buys by splitting the data finer. The within-block control adds +0.000.

And it is ONE BIT, not a count. Replacing n_sym_root by the bare flag "no root production survives every block at once" keeps a median 100% of the gain; grading the surviving count adds 0% on top -- and that is not because the count had no room to grade: it is >= 2 at a median 12% of observations. So the straddling window is running a CONSISTENCY CHECK, which is what hshunt asked about, and not accumulating evidence.

Unregistered and worth flagging: the SIGN of the effect on k flips with v. At v=2 a root-tightened site gets a SHORTER window (6/6 cells); at v=4 a LONGER one (6/6). I registered no direction here and am not claiming a mechanism -- the last two mechanisms I stated were both refuted while the predictions held.

6. The bit survives the exact posterior -- and so does the null





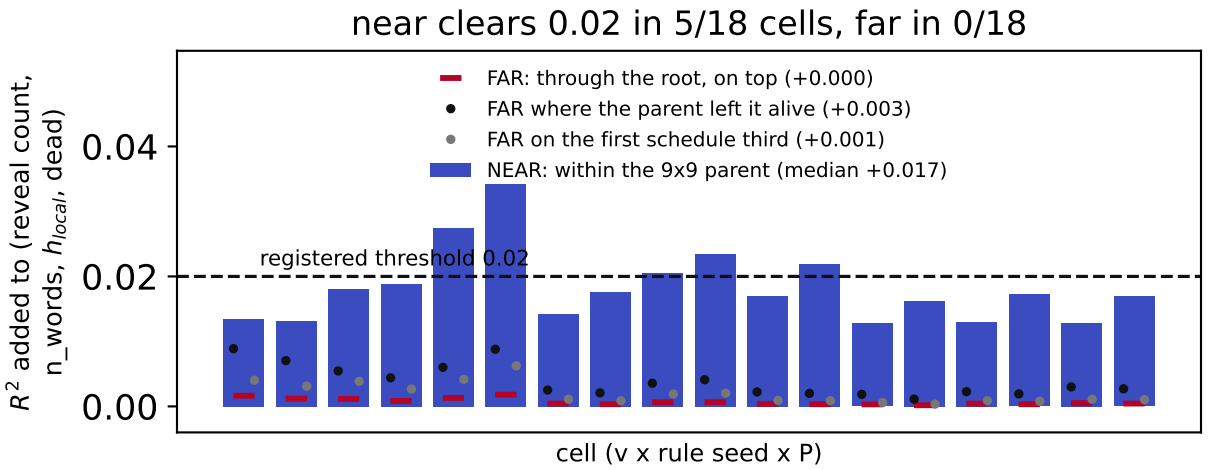
Page 5 rested on a COUNT, which only asks which symbols are possible, never how probable. BP is exact on this tree, so the graded version is closed-form: h_root = H(block symbol | ALL observed leaves). If the criterion tracked anything softer than a yes/no consistency check, that is where it would show up.

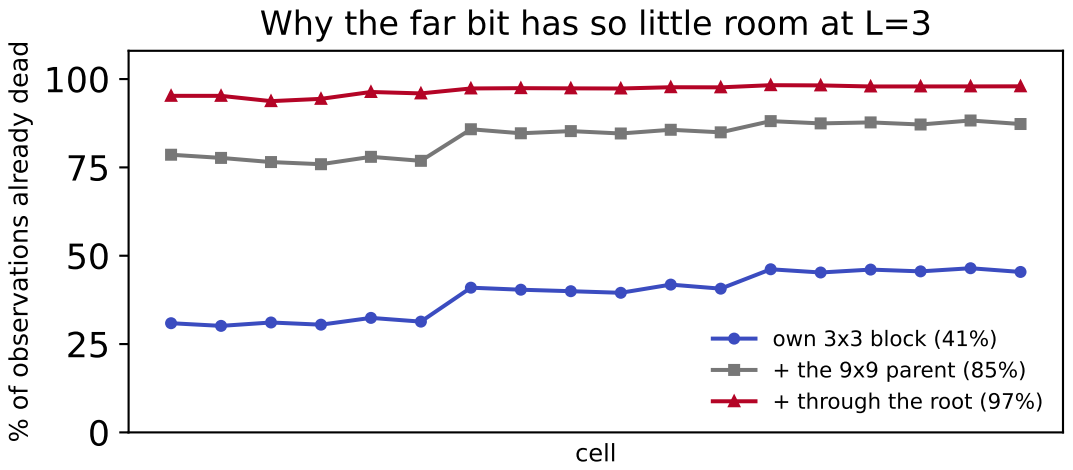
It does not. On top of (reveal count, n_words, h_local, dead) the exact posterior gains +0.000 in 0/18 cells, net -0.000 of its null. The bit itself re-earns +0.197 [+0.116..+0.294], reproducing page 5 to three decimals from an independent computation -- and BP deadness equals the enumeration predicate at 1.0000 in 18/18.

The feature had room to grade: 6-9 bins over 33-218 distinct values, the zero atom (root pins the symbol) at 63-80% of live observations. So this null is a much harder statement than the count's was: the BAYES-OPTIMAL posterior adds nothing over one bit. That was the outcome I registered as the stronger of the two.

Unregistered, same BP pass: h_leaf, the exact posterior entropy at the SITE's own token -- the quantity you named for this criterion. It explains 0.416 of k alone and adds +0.000 over the baseline. The criterion behaves CONSISTENTLY WITH tracking a posterior entropy, while that entropy holds nothing the reveal count, the block's word count and the bit do not already hold.

7. At depth: the check reaches its own parent, not the root





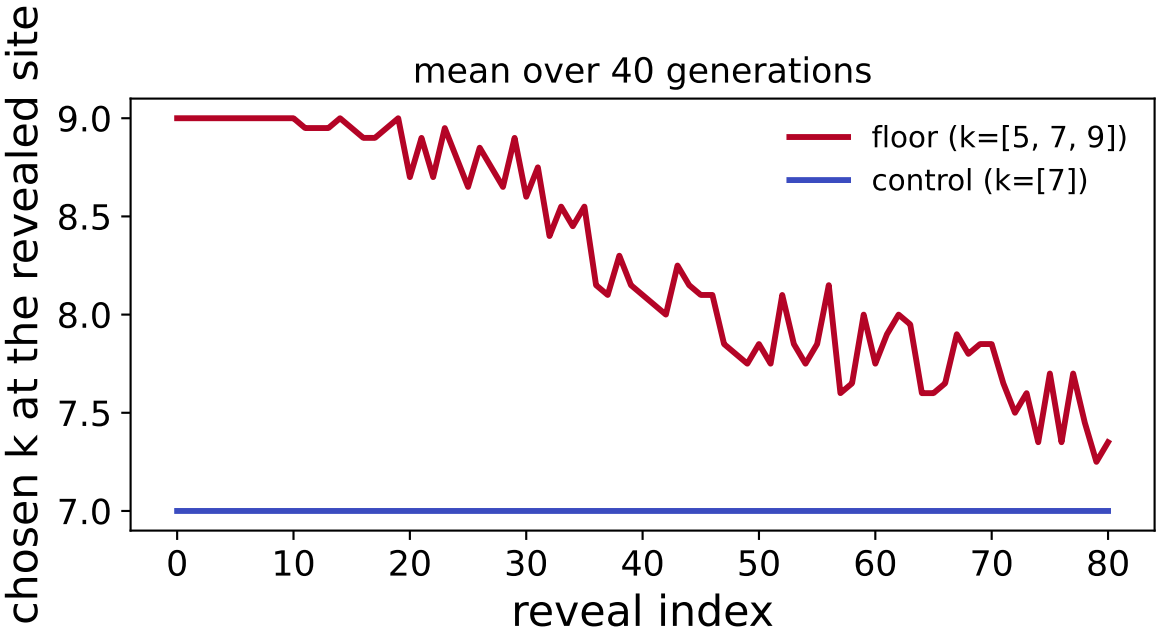
Every cross-block result so far was read at s=3, L=2 -- where a block's only parent IS the root, so "consistent with the other blocks" and "consistent through the root" are the same sentence. L=3 is the first board that splits them: the parent is a 9x9 region, the root couples nine of those. Free check first -- at L=2 the two messages agree to 0.0e+00 and the near gain reproduces page 5's bit at +0.218, from code that shares nothing with it.

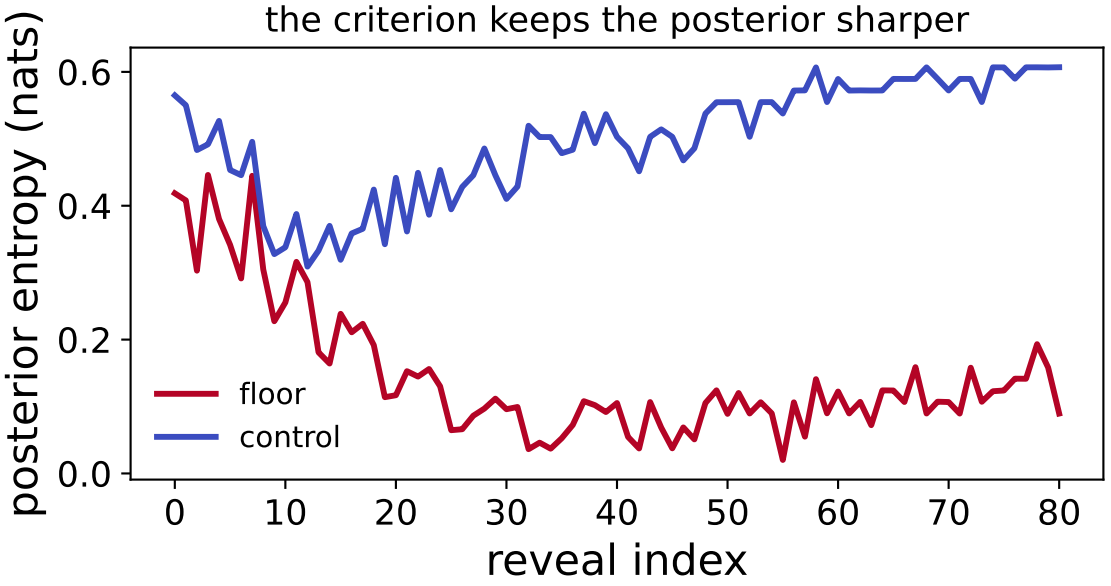
At L=3 they separate, and the answer is near. The parent adds +0.017 (10% of the residual); the root adds +0.0005 on top of it, 37x smaller. The two cuts built to give the far bit its best shot agree: +0.0029 among blocks the parent left alive, +0.0011 on the first third of the schedule.

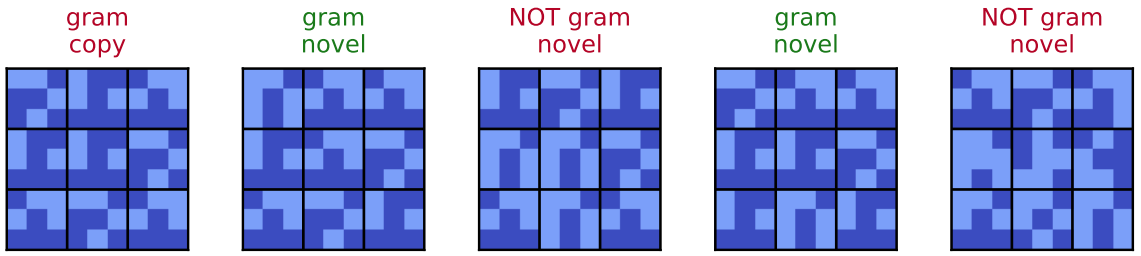
The honest caveat, measured before the run and not tuned away: at L=3 a grid is 729 sites, so P* = exp(Theta(n)) is astronomically past the P <= 1000 here. The sampler goes off-grammar early -- dead at the root for 97% of observations, already 95% in the FIRST third of the reveal schedule, and worse at larger v (95% at v=2, 98% at v=4), so no setting buys that bit range. Items 6 and 7 exist because of it: only 15% of observations have a parent that has not already killed them, and that is where the root bit still splits. It adds nothing there either.

So the consistency check hshunt described is real but LOCAL: a straddling window checks its own enclosing block, and the L=2 result was that near coupling all along, read at the one depth where near and far coincide.

8. Example traces: the window size moving along one generation







criterion output, the FIRST 5 generations (not selected)
v2 m2 rs0, P=300. Black lines mark the 3x3 blocks.

Same grammar, training set, sampler and seed in both arms; only the window rule differs. The criterion starts at k=9 and backs off as the grid fills; the control is pinned at the best fixed k=7 for the whole run.

	success	gram	novel
criterion	0.4062	0.6250	0.7812
fixed k=7	0.0625	0.0938	0.9688

The win is grammaticality (6.7x here), bought with a little novelty: the fixed window is novel almost always because it is mostly emitting grids that no root derives.

Where this leaves the patch-size question

what was tried	effect	support	reading
choose the window adaptively vs any fixed k	12x	12/12 cells	the length scale is the whole story
add hshunt's entropy floor h_min > 0	worse	12/12 cells	h_min=0 (bare support) is best everywhere
add the count floor n_min > 1	worse	both P rungs	same mechanism: rejection shortens the window
replace per-site choice with its own k(t)	15% kept	9/9 cells	at matched mean k -- same budget, worse spent
predict k from position (block offset)	R^2<0.001	18/18 runs	no positional signature at all
predict k from local reveal count	R^2<=0.47	18/18 runs	subsumes time; rest is content-dependent
name that residual: within-block grammar	+0.107 R^2	18/18 cells	n_words, over the reveal count
and grammar ACROSS block boundaries	+0.196 R^2	18/18 cells	net of a permutation null; hshunt's straddle
grade it: exact BP posterior, not a count	+0.000 R^2	0/18 cells	it is one consistency bit, not evidence
posterior entropy at the site itself	+0.000 R^2	18/18 cells	explains 0.42 alone, nothing once conditioned
how FAR does that consistency reach (L=3)	+0.017 near	5/18 cells	its own enclosing 9x9 block, 10% of residual
the same bit through the ROOT, on top	+0.0005 far	0/18 cells	null on 3 independent cuts; 37x under near

Answer to "how are you setting the patch size over time?"

Not over time. The rule that works is "the largest window that matches any training patch at all", evaluated SEPARATELY AT EVERY SITE. Its k(t) curve is a summary of what it does, not a specification of it: handing that same curve back to the sampler recovers only 15% of the success.

The entropy criterion in particular does not bind -- every floor above zero costs success in every cell measured, because a rejected window falls back on a shorter one rather than on a better one.

What the site-dependence IS, is grammar: the block's own ambiguity, and then its consistency with the other blocks inside its own enclosing 9x9 region -- L=3 says the root itself adds nothing on top (page 7) -- which only a window straddling a boundary can see. Page 5 counts it; page 6 redoes it with the exact BP posterior and finds the same thing -- it is ONE consistency bit, and the site's own posterior entropy adds nothing once that bit is conditioned on.

Caveat: the sweeps are s=3, L=2 (9x9, rungs 5/7/9), v in {2,3,4}, m=2 mostly, n_gen=4096; the depth test is s=3, L=3 (27x27, rungs 5/11/17), n_gen=64 matched on n_obs. Deeper boards need k >= U1 = 2s^(L-1)-1, out of reach at L=4. At L=3 the sampler is far below P*, so the far bit is read where it is already violated: the null is checked on 3 cuts (marginal, parent-alive, early), not asserted from one.